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ORIGINAL INVESTIGATION

Influence of Travel Demands and Match Load on Recovery Following Postmatch Travel in National-Team Footballers

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Purpose: This study investigated the relationship between travel demands and match loads on perceptual recovery, fatigue, and sleep following postmatch travel in national football teams. In addition, the influence of travel demands and the time between match kickoff and travel departure on postmatch recovery was examined. ***Methods:*** Match-running load (via GPS) and travel data were obtained from 79 male national-team footballers. Postmatch travel duration, direction, context, time-zone difference, and time between kickoff and travel departure were collated. Athletes provided perceptual ratings of fatigue, soreness, sleep, stress, and recovery from 1 day prematch through to 3 days after postmatch travel. Linear mixed models assessed the influence of match load and travel on perceptual ratings for 3 days posttravel. Additional models assessed a standardized postmatch time point of match day + 3 to determine whether timing and extent of travel influenced recovery. ***Results:*** Higher match loads were associated with poorer recovery, fatigue, soreness, and sleep ($P = .001–.032$). Athletes reported poorer fatigue, soreness, and recovery when traveling from national teams back to clubs compared with between national-team matches ($P < .001$). Traveling eastward was associated with poorer sleep ($P = .004$). Longer periods between kickoff and travel departure were associated with poorer sleep on match day + 3 ($P = .003$). ***Conclusions:*** Perceptual recovery, fatigue, sleep, and soreness following postmatch travel were affected by both match load and travel demands. Greater match loads and eastward postmatch travel may impair recovery. In addition, departing later after a match was associated with poorer sleep on match day + 3; however, no relationships existed to any other recovery measures.

Keywords: jet lag, match recovery, fatigue, football, transition

National football (soccer) team athletes are often required to travel back to their clubs shortly after the completion of a national-team match and then be prepared to compete again for their club team within 3 days.[1] Hence, postmatch travel is a concern for national football teams due to the physiological and physical stresses of travel combined with the need to recover from match play.[1–3] Furthermore, travel stressors and the time required for travel may restrict time available to undertake postmatch recovery and affect recovery status. The combined effects of both match load and travel requirements are likely to have significant implications on fatigue, sleep, and recovery, potentially also impacting selection for their next club match.[3,4] Understanding how match load and postmatch travel influence perceived recovery, fatigue, sleep, and soreness can assist practitioners to plan for travel after both international and club matches.

International football windows are typically 9-day periods in which club football teams must release their athletes to national teams to compete in up to 2 international matches. Within these 9-day windows, players travel from clubs to national teams (often immediately after club matches), compete in 2 international matches with transition travel in between, and then return for their next club team fixture within a few days. Due to the condensed nature of these windows, athletes are often afforded only a few days to recover from both match loads and subsequent travel demands. Currently, very little research explores the combined effects of these 2 stresses on postmatch recovery to guide subsequent player availability in any sport but, particularly, for football. Kraemer et al[5] observed elevated blood recovery markers following a simulated team-sport match and 5 hours eastward trip in physically trained individuals; however, as both the travel and match load were the same for all participants, the influence of variations in these variables is unknown. Studies examining short-haul travel in Australian domestic football have reported no difference in postmatch recovery[6] or perceptual wellness[7] between home and away matches, though they only incorporated short-haul travel in a single club team. Furthermore, the aforementioned studies address only a fraction of the possible travel demands a national team footballer may experience, and no studies compare the recovery responses of athletes across a broad range of travel demands. Although evidence exists that national team footballers report poorer perceptual fatigue, sleep, and soreness following travel out of a national team compared with travel into or between matches,[8] the specific effects of different match loads and postmatch travel demands on recovery are unknown. Detailed understanding of player responses to combined match and travel loads is, therefore, needed to better inform player availability and recovery when traveling between national team matches and when returning to club teams.

The need to travel to subsequent national team matches or return to club teams shortly after demanding national team matches may create suboptimal conditions for recovery, which may influence preparation and selection in subsequent matches. Although the timeline of recovery from a professional football match is reported to range from 5 to 96 hours,[9] the conditions of travel are often suggested to interfere with the recovery of athletes following matches and prolong this timeline. Although a common occurrence in modern fixtures, this is not ideal when only 3 to 5 days separate

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matches and include extensive travel. Despite this major concern, no research has investigated how player recovery differs based on postmatch travel demands. A recent review on postmatch recovery strategies highlighted sleep and nutrition as being particularly important for match recovery.[10] Postmatch travel, however, has been previously observed to reduce total sleep in Rugby 7's players,[11] and reductions in sleep during and after travel have frequently been reported by traveling athletes.[12–15] As such, it is possible that the postmatch recovery of athletes may be impaired by travel-induced sleep loss. In addition, travel demands may also reduce time available for other recovery strategies, that is, nutrition, cold-water immersion, or massage, which have also been suggested to have beneficial effects on perceptual recovery.[10] Due to the potential impacts of travel on postmatch recovery, practitioners of national teams are also left to consider the best time to travel after a match. However, it is unclear whether traveling later and spending time recovering prior to travel will eventuate in better outcomes than traveling immediately postmatch and recovering after travel. Given that an athlete may be required to compete again within 3 days of their previous match, understanding how the timing and extent of travel influence postmatch recovery will help inform the planning of travel schedules (eg, when to depart) and guide posttravel recovery plans.

Based on the lack of research surrounding the relationships between match load and travel, this study has 2 aims. First, to describe the relationship of match load and travel duration, time zone change, and direction on the perceptual fatigue, sleep, soreness, and recovery status of national team footballers upon arrival from postmatch travel and, second, to describe the influence of travel duration, time zone change, direction, and time between match and travel departure on match recovery status 3 days after a match in national team footballers. By addressing these 2 aims, this study will provide staff of club and national teams better insight on the expected recovery response of players as related to their match load and travel demands.

Methods

Subjects

Participants included 79 male international footballers (age = 25.00 [3.67]) competing for a national football team between October 2018 and November 2023. Data were collected from each trip that the players undertook for either the senior or under 23s national teams (n = 258). Through contracts with the national football federation, all players provided consent for their data to be used anonymously for research purposes. Ethics approval was obtained from the University of Technology Sydney's human ethics committee (ETH22-7708).

Methodology

Data were collected from each national team player traveling after a national team match (including travel both between national team matches and back to clubs). Participants were required to complete a perceptual questionnaire every day when in the national team camp and up to 3 days following return to their club. Players were also required to provide a perceived recovery status rating from the day prior to each match through to 3 days after the match. Each player's perceptual scores were aligned with their match loads and postmatch travel details. To accomplish aim 2, perceptual outcome scores were collated from a standardized postmatch time point of match day + 3 (MD + 3). This time point was selected as all players had completed postmatch travel requirements by this point but allowed for a standardized time point for comparison that could align with the earliest possible club match. Baseline scores for both aims 1 and 2 were obtained the morning before a match (MD). Perceptual outcomes were converted to change scores by subtracting the baseline score from the raw value.

Travel Details

Travel details for each postmatch flight were obtained from booked travel schedules provided by national team managers. Actual departure and arrival times for each flight were verified via an online flight database (Flightera.com). Travel duration (mean: 14.51 [4.10] h, range: 1.08–32.98 h) was calculated as the total time between airport departure and arrival at the final destination airport (including the duration of stopovers but not travel to and from airport). The recovery time between matches and travel was calculated as the length of time in hours between match kickoff and aircraft departure due to the availability of match start time (mean: 24.34 [12.11] h, range: 4.83–76.33 h). The number of time zones crossed during travel was measured as the difference in time zone between departure and destination locations at the time of travel (mean: 5.56 [2.25], range: 0–14). The direction of the flight was labeled as either eastward or westward based on the flight path. The travel context was categorized as being either transition (travel between 2 national team matches) or return (travel out of national team back to club team). The travel variables used have been explored in prior works where they have been shown to have a relationship with posttravel perceptual fatigue and sleep measures.[8,16] Athletes stayed in team hotels during camp periods (ie, after transition travel), whereas following return travel, athletes typically stayed in their own accommodation at their respective club teams.

Match Load

GPS devices measured the running loads of players during each match. Players wore a GPS unit (STATSports, Apex) worn between the players' scapulae using a customized vest. Each unit included a gyroscope (10 Hz), triaxial accelerometer (100 Hz), and magnetometer (10 Hz). Satellite quality was filtered for using HDOP values of <1. These devices have previously been validated for the measurement of speed and distance variables in team sports.[17,18] The following measures were obtained for each player from each match: total distance, total high-speed distance (19.8–25.2 $km \cdot h^{-1}$), total very high-speed distance (VHSD; >25.2 $km \cdot h^{-1}$), number of accelerations (>3 $m \cdot s^2$), and number of decelerations (>3 $m \cdot s^2$).

Perceptual Measures

Players completed a perceptual monitoring questionnaire every morning during the national team camp and for 3 days after the completion of the camp. The questionnaire included 4 subjective ratings on a 7-point Likert scale. Players were asked to rate their current level of fatigue, soreness, sleep quality, and stress as used in prior works.[16,19] A "Total Wellness" score was calculated from the sum of each of the individual scales. Athletes were also asked to complete the Perceived Recovery Status Scale[20] each day from the day before a match through to 3 days after the match. This scale was a 10-point scale requiring players to rate their current perception of their recovery status, with scores of 10 relating to being very well recovered and scores of 0 relating to being very poorly recovered. Athletes completed all perceptual monitoring scales

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via the federation's athlete monitoring software (Smartabase Athlete, Fusion Sport Pty Ltd) installed on their own smartphone.

Although the authors acknowledge that the validity of perceptual wellness scales to monitor athletes can be questioned,[21] these scales have previously been observed to be responsive to variations in match load.[22–24] In addition, these scales have been found to be responsive to variations in travel demands, including travel duration, time zone difference, and travel direction.[8,16] Given the challenges associated with athlete monitoring in national football teams,[25] these tools represent a practical way to monitor athlete responses to both match and travel stresses.

Statistical Analysis

All perceptual monitoring scales were aligned with the prior travel details and match GPS measures in an Excel spreadsheet. For aim 1, perceptual scores were labeled as either D1, D2, or D3 (days 1–3, respectively) based on the timing of the score relative to arrival from travel. All data were imported into R Studio (version: 2023.6.0.421) for analysis. Linear mixed models were created for each perceptual outcome variable.[26] Models were built using a forward step-wise approach with the inclusion of each fixed effect determined by statistical significance ($P < .05$), measured via F tests with Satterthwaite degrees of freedom approximations using the lmertest package.[27] Where multiple effects were deemed statistically significant, the Akaike information criterion value and the R^2 values of the model were used to determine order. Where fixed effects were correlated (ie, all match load variables and travel duration/time zone difference), only one of the strongest of the 2 variables was included to control for multicollinearity within models. The following fixed effects were tested for each model: day, travel duration, time zone difference, travel direction, travel context, match total distance (in meters), match high-speed distance, match VHSD, match accelerations, and match decelerations. All players were assigned an anonymous identification number, which was included as a random effect within the model to account for variation between participants. The national team camp that the data came from was also coded in as a random effect.

For aim 2, perceptual measures taken on MD + 3 were used as outcomes for separate linear mixed models. Models were built using the same methods described for aim 1. Models were built using the same fixed and random effects as those listed earlier, with the additional included fixed effect of the recovery time.

Results

Model Outcomes

Details of the final selected models for each outcome are presented in Table 1. Overall, the model R^2 values showed fatigue (R^2 marginal = .21) and recovery (R^2 marginal = .22) scales to be the most responsive to combined travel and match stresses.

Posttravel Perceptual Outcomes

Athletes reported poorer ratings of fatigue (0.491 AU increase; $P < .001$), soreness (0.297 AU increase; $P < .001$), total wellness (0.529 AU increase; $P = .007$), and recovery (0.629 AU decrease; $P < .001$) following return travel than when they traveled between national team matches (Figure 1). Traveling eastward resulted in significantly poorer sleep responses than traveling westward (0.429 AU increase; $P = .004$), and regardless of travel context, traveling later after the match also had a negative effect on sleep ratings 3 days following arrival (0.014 AU increase per hour later; $P = .003$; Figure 2). No significant effects existed for any other travel variable on perceptual markers ($P > .05$).

The strongest associated match load measure differed between perceptual outcome variables. Increases in match VHSD had a negative effect on sleep (0.003 AU increase per meter VHSD; $P < .001$) and recovery (0.002 AU decrease per meter VHSD; $P = .032$) scores. A greater number of decelerations in matches resulted in poorer ratings of fatigue (0.002 AU increase per deceleration; $P = .031$), soreness (0.002 AU increase per deceleration; $P = .023$), and total wellness (0.012 AU increase per deceleration; $P < .001$). Greater match accelerations were also associated with poorer stress ratings (0.002 AU increase per acceleration; $P = .008$).

MD + 3 Perceptual Outcomes

Final model outcomes for MD + 3 perceptual measures are shown in Table 2. Traveling eastward after matches was associated with poorer fatigue (0.432 AU increase; $P = .002$), sleep (0.502 AU increase; $P = .032$), total wellness (1.030 AU increase; $P = .015$), and recovery (0.853 AU decrease; $P < .001$) ratings when compared with traveling westward. Athletes also reported poorer sleep (0.024 AU increase per hour later; $P = .003$) and total wellness scores (0.032 AU increase per hour later; $P = .023$) the later after the match they traveled (Figure 3). Perceptual recovery ratings were worse when traveling for longer durations (0.065 AU decrease per hour traveled; $P = .003$). At MD + 3, greater VHSD in matches was associated with poorer sleep (0.003 AU increase per meter VHSD; $P = .019$) and total wellness scores (0.005 AU increase per meter VHSD; $P = .030$). No other effects of match load were observed at MD + 3.

Discussion

The current study explored the relationship of match load and travel demands on perceptual fatigue, sleep, and recovery responses of professional footballers following national team matches. Perceptual ratings of recovery, fatigue, soreness, and sleep were influenced by both match external load and the postmatch travel demands. Greater match acceleration, deceleration, and high-speed running load was associated with poorer recovery, fatigue, soreness, and sleep, with these measures also reported to be worse following travel back to club compared with travel between national team matches. In addition, players who traveled eastward after matches reported poorer sleep and recovery than those traveling westward. Furthermore, players with longer travel durations reported poorer perceptual recovery on MD + 3, and players who traveled later after a match reported poorer sleep on MD + 3 than those who traveled earlier.

The findings of this study indicate that the perceptual fatigue, sleep, and recovery state of national team footballers upon arrival from postmatch travel showed small associations (R^2 = .02–.22, as context an R^2 of 1.0 represents that all variation in the outcome value is explained by the model and its fixed effects) with high-speed running loads undertaken in the prior match and the travel demands after matches. Greater high-speed loads (measured as either high-speed distance, VHSD, or acceleration/deceleration counts) were negatively associated with all perceptual recovery measures. These findings concur with recent works in both senior[22] and junior[23] professional footballers reporting negative

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Table 1 Model Outputs for Changes in Wellness and Recovery Scales From Baseline (MD or MD − 1) and Posttravel (Arrival + 1, +2, +3)

Outcome	R^2 M	R^2 C	Explanatory variable	Coefficient	*P*
Fatigue (n = 258)	.21	.33	Intercept	0.555	
			D2	−0.485	<.001
			D3	−0.656	<.001
			Trip type: Return	0.491	<.001
			Match decelerations	0.002	.031
Soreness (n = 258)	.09	.21	Intercept	0.405	
			D2	−0.248	.001
			D3	−0.453	<.001
			Trip type: Return	0.297	<.001
			Match decelerations	0.002	.023
Sleep (n = 258)	.10	.29	Intercept	−0.070	
			Match VHSD	0.003	<.001
			D2	−0.351	.002
			D3	−0.430	.001
			Travel direction: East	0.429	.004
			Recovery time	0.014	.003
Stress (n = 258)	.02	.21	Intercept	0.015	
			Match accelerations	0.002	.008
Total (n = 258)	.13	.32	Intercept	1.482	
			D2	−1.074	<.001
			D3	−1.590	<.001
			Match decelerations	0.012	<.001
			Trip type: Return	0.529	.007
Recovery (n = 181)	.22	.40	Intercept	−0.890	
			D2	0.667	<.001
			D3	1.086	<.001
			Trip type: Return	−0.629	<.001
			Match VHSD	−0.002	.032

Abbreviations: MD, match day; R^2 C, R^2 conditional; R^2 M, R^2 marginal; VHSD, very high-speed distance. Note: Coefficient values represent change in outcome variable per unit change in explanatory variable (1 for categorical variables).

associations between increased high-intensity running and perceptual fatigue, soreness, energy, and sleep on MD + 1[22] and MD + 2,[23] respectively. However, similar to the findings in a recent systematic review on perceptual responses to training and match load,[28] the variation in scores explained by these models was relatively low (R^2conditional = .04–.22), indicating that other factors are also likely to influence perceptual scores. Regardless, this study is the first to show that pretravel match loads, specifically high-speed loads, continue to influence an athlete's perceptual recovery status following arrival from postmatch travel. As such, high-speed match loads should be considered when planning for the expected arrival status of athletes. Furthermore, these findings emphasize the importance of sharing match load information between clubs and national teams to better inform athlete status and availability upon return to clubs.

In addition to the influence of match load, both postmatch travel direction and the context of travel appeared to influence an athlete's postarrival state. Athletes traveling eastward after a match reported poorer sleep scores (0.429 AU increase) postarrival compared with those traveling westward. This finding concurs with Fowler et al,[15] who identified reductions in sleep duration and time in bed following eastward travel across 8 time zones compared with westward travel of the same extent. The reduction in sleep quality in this study may be explained by delayed sleep onset as a result of jet lag, with morning commitments potentially limiting the capability to make up for the delayed onset of sleep, though we recognize that neither sleep onset nor wake time was recorded in this study. As such, athletes with prolonged travel eastward, especially overnight after a national team match, may require additional time to sleep or strategies to improve sleep quality following arrival. Travel context, as either transition between national team matches or returning to club, also appeared to influence perceptual state upon arrival. Perceptual status was poorer following travel back to club teams, with an increase in fatigue (0.491 AU), increase in soreness (0.297 AU), and decrease in recovery ratings (0.629 AU) observed in athletes returning to clubs compared with those traveling between national team matches. This finding has previously been reported in the same population group using a smaller sample size, showing that perceptual fatigue, sleep, soreness, and total wellness scores up to 3 days posttravel were lower following travel out of national teams compared with travel into national teams or between national team

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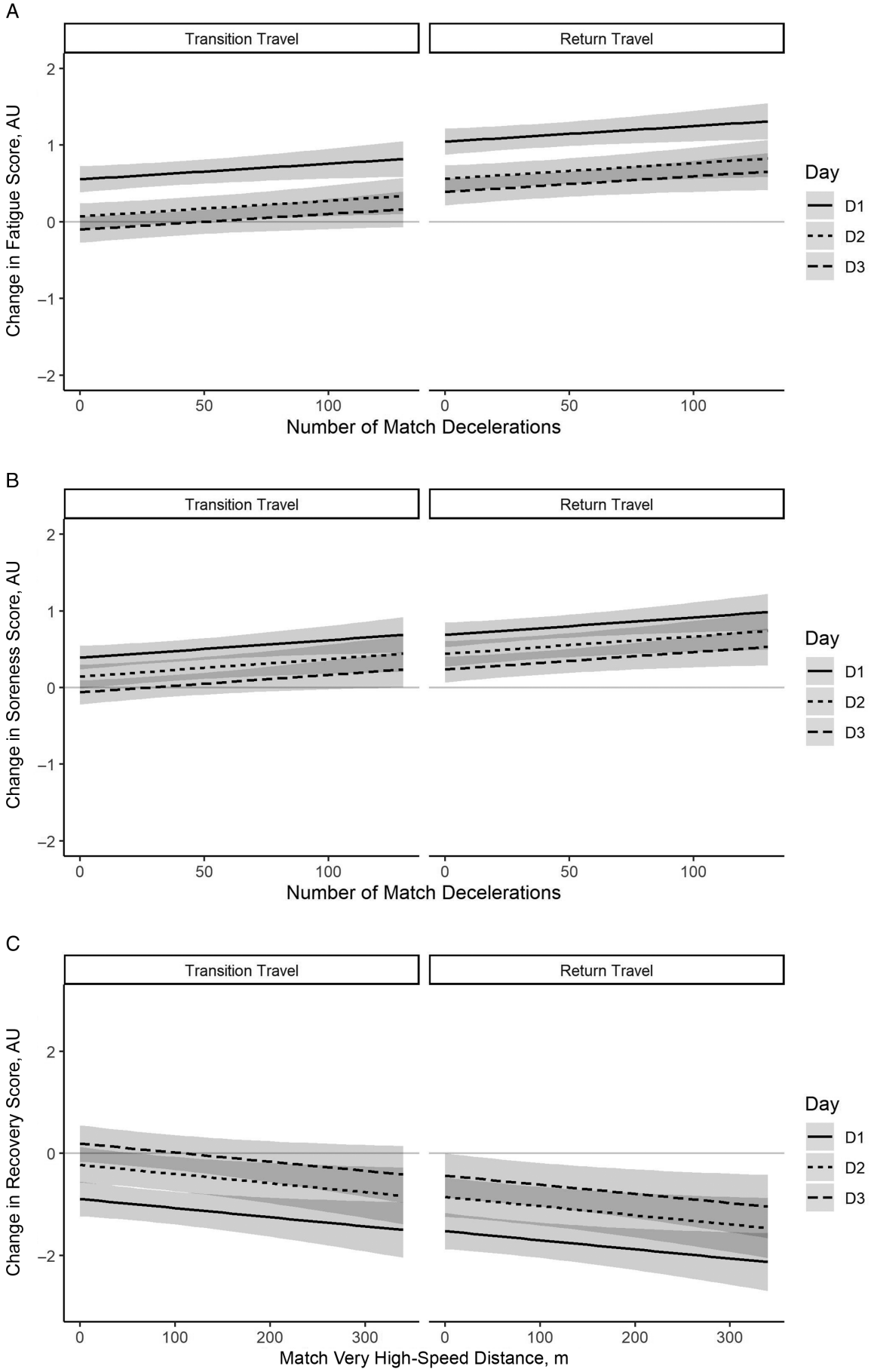

Figure 1 — Change in (A) fatigue, (B) soreness, and (C) recovery scores between prematch to posttravel based on travel type, match decelerations, and match very high-speed distance. Shaded areas represent the 95% CIs.

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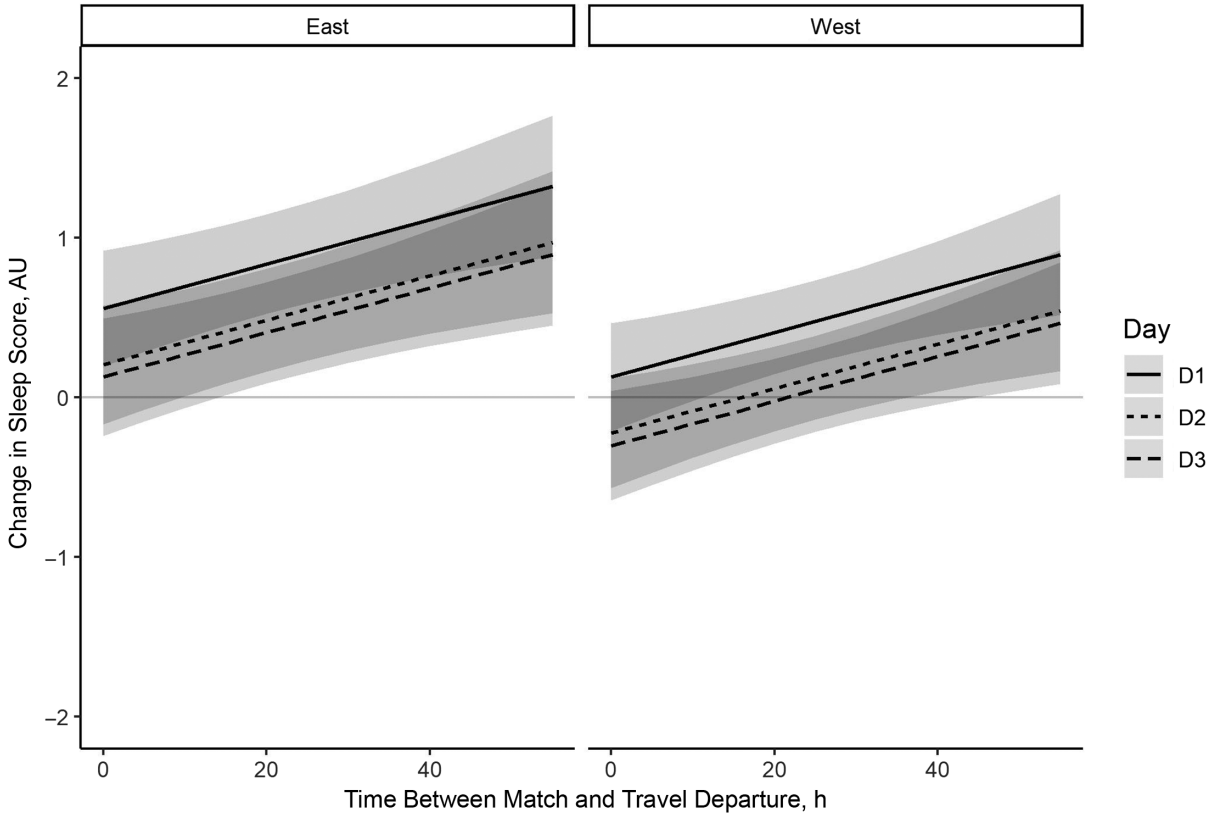

Figure 2 — Change in sleep scores from prematch to posttravel based on the time between match kickoff and travel departure and the travel direction. Shaded areas represent the 95% CIs.

Table 2 Model Outputs for the Difference in Wellness and Recovery Scales From Baseline (MD or MD − 1) and 3 Days Postmatch (MD + 3)

Outcome	R^2 M	R^2 C	Explanatory variable	Coefficient	*P*
Fatigue (n = 228)	.05	.22	Intercept	0.525	
			Travel direction: East	0.432	.002
Soreness (n = 228)	.00	.09	Intercept	0.404	
Sleep (n = 228)	.09	.22	Intercept	−0.333	
			Recovery time	0.024	.003
			Travel direction: East	0.502	.032
			Match VHSD	0.003	.019
Stress (n = 228)	.00	.22	Intercept	0.083	
Total (n = 228)	.08	.35	Intercept	0.368	
			Travel direction: East	1.030	.015
			Recovery time	0.032	.023
			Match VHSD	0.005	.030
Recovery (n = 168)	.15	.36	Intercept	−0.204	
			Travel direction: East	−0.853	<.001
			Travel duration	−0.065	.003

Abbreviations: MD, match day; R^2 C, R^2 conditional; R^2 M, R^2 marginal; VHSD, very high-speed distance. Note: Coefficient values represent change in outcome variable per unit change in explanatory variable (1 for categorical variables).

matches.[8] This may relate to athletes returning to clubs at the end of a national team camp having experienced multiple national team matches and travel bouts in a relatively short period of time. Accordingly, for players transitioning back to club teams from national teams, additional strategies to maintain sleep during travel are highly recommended,[29] and recovery interventions that can be implemented during periods of travel may further benefit the athlete.[10]

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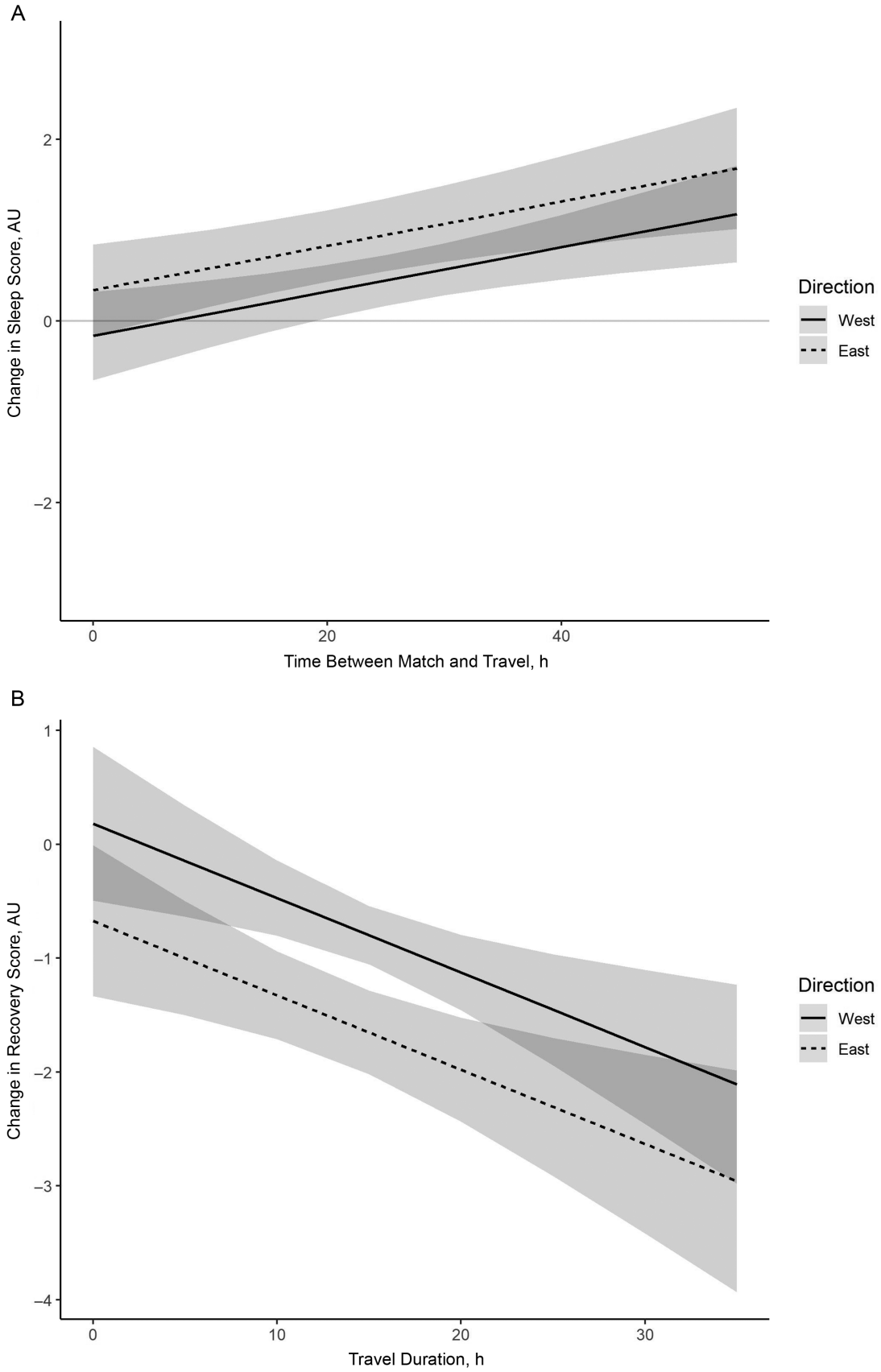

Figure 3 — Change in sleep and recovery scores from match day and match day + 3 based on travel direction, the time between match kickoff and travel departure, and the duration of travel. Shaded areas represent the 95% CIs.

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Due to the potential need to compete again within 3 days after national team matches, a secondary aim of this study explored how long after a match travel occurred and effects on postmatch recovery 3 days after a national team match. Longer travel demands in this study appeared to affect perceptual recovery from matches, with athletes reporting a decrease in perceptual recovery scores on MD + 3 (0.065AU) for every hour traveled after matches. This is a novel finding as no other studies have compared an athlete's postmatch recovery at a standardized time point based on differing postmatch travel requirements, recognizing that players travel at different times following the match. Poorer recovery ratings when required to travel longer durations may, in part, be associated with reductions in sleep duration and quality that have been frequently reported during long-haul travel.[12–15] However, as sleep duration or quality during travel was not recorded in this study, further research is necessary. In addition to the duration of travel, athletes traveling eastward after matches reported on MD + 3 an increase in fatigue (0.432 AU), poorer sleep (0.502 AU), and decrease in recovery ratings (0.853 AU) compared with those who traveled westward. As discussed previously, eastward travel is likely to result in delayed sleep onset caused by jet lag, which has previously been observed to reduce sleep duration.[15] Reductions in sleep because of this may further exacerbate feelings of fatigue from matches and impair postmatch perceived recovery status. Based on the findings of this study, it appears that longer postmatch travel durations, particularly in an eastward direction, may impair recovery from national team match play and result in poorer ratings of recovery 3 days after a match. This poses concern for athletes who are required to compete within 3 days after their national team match, and recovery tools that can be applied during travel are recommended.

A concern among practitioners of national football teams often revolves around whether it is better to travel immediately after a match or wait and allow athletes time to recover from the match prior to traveling. Athletes in this study reported an increase in sleep score (0.024 AU) on MD + 3 per hour later that they traveled after the match, whereas the timing of travel had no effect on any other perceptual recovery measures. Better sleep scores on MD + 3 when departing earlier after a match may be related to having more time to adapt to the destination time zone and recover from sleep deficits associated with long-haul travel. As such, if athletes are required to compete again shortly after a match, traveling earlier may allow adaptation of sleep schedules to occur earlier. However, as this is the first study to explore the relationship between the timing of travel relative to a match and match recovery, further research is needed to support this finding.

Although the findings of this study may inform practitioners of the travel and match load influence on perceptual recovery after national team matches, several limitations should be considered. The population used within this study represented only a single national team, and thus, variation may be expected from other national teams, which are likely to have their own unique travel demands and strategies. Likewise, the speed and acceleration thresholds used for this research represent those employed by the studied national team, and variations in outcomes may be expected where different thresholds are used. In addition, as evidenced by low R^2 values, the variation in perceptual recovery measures used in this study is only explained to a small extent by travel and match-related factors. Other influences on perceptual recovery from matches and travel are likely to exist and should be considered when monitoring athletes traveling postmatch. Furthermore, this research only considers air travel requirements as data were not available for other forms of transport. Finally, the design of this study meant that the research team had no control over what the athletes did before, during, or after travel or over the use of recovery strategies postmatch.

Practical Applications

- Elevated high-speed match-running demands are likely to affect perceptual fatigue, sleep, and soreness ratings following travel after national team matches.
- Athletes may require additional support following travel back to clubs after national team matches due to poorer perceptual recovery, fatigue, and soreness compared with travel between national team matches.
- Perceptual recovery on MD + 3 following travel is negatively impacted by longer postmatch travel durations, and traveling later after the match may have negative impacts on sleep on MD + 3.

Conclusions

In conclusion, this study was the first to determine the combined influence of travel and match demands on perceptual recovery, fatigue, and sleep following postmatch travel. Increased high-speed match running in the prior national-team match was associated with poorer perceptual recovery, fatigue, sleep, and soreness on arrival from travel. In addition, traveling from the national team back to the club was associated with poorer ratings of recovery, fatigue, and soreness compared with traveling between national-team matches, and traveling eastward was associated with poorer sleep ratings. In addition, the perceptual recovery status on day 3 after a national-team match appeared to be negatively affected by longer postmatch travel demands. Finally, traveling later after a match was associated with poorer sleep ratings at MD + 3. When monitoring athletes traveling after national-team matches, practitioners should be aware of both prior match loads and travel demands of their athletes. In addition, for athletes who are required to compete again within 3 days of their prior national-team match, travel-friendly recovery strategies are recommended to maximize postmatch recovery.

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